

Topic 3.4 – Future of Food

“In the past year, we have seen food riots on three continents, food inflation has rocketed and experts predict that by 2050, if things don’t change, we will see mass starvation across the world.” – BBC, ‘The Future of Food’

Food is both a celebrated and contested issue. It is predicted that 805 million people go to bed hungry each night, while others consume and waste far more than their fair share. Across the planet food security varies both within and between countries at all levels of the development spectrum.

This topic explores the spatial patterns and complex causes of food security, from the physical influences on food systems and how humans create and exacerbate food security issues. Learners will investigate the impacts of food systems on people and the environment before considering management strategies at a range of scales, including an in-depth case study of one country’s efforts to improve food security.

1. What is food security and why is it of global significance?	
Key Ideas	Content
1.a. The concept of food security is complex and patterns of food security varies spatially.	- Defining what it means to be food secure and understanding that the concept of food security is built on three pillars of food access, food availability and food use. - Current trends in global food security using data such as undernourishment and hunger statistics and the Global Food Security Index. - How the pattern of food security is dynamic and varies both between and within countries.
1.b. Food is a precious resource and global food production can be viewed as an interconnected system.	- The physical conditions required for growing food including, air, climate, soil and water. - How feeding the world is a complex system of growing, processing, transporting and disposing of consumer waste. - How food production methods vary from intensive to extensive and subsistence to commercial.
1.c. Globalisation is changing the food industry.	- The influence of globalisation on the food industry such as increased demand and global tastes. - Globalisation of the food industry creates a number of issues including food miles, inequality between TNCs and small suppliers, obesity and price crisis. - Globalisation of the food industry creates a number of opportunities including technological innovation, short-term food relief and consumer choice.

2. What are the causes of inequality in global food security?

Key Ideas	Content
2.a. A number of interrelated factors can influence food security.	• Understand the range of physical factors that affect food security across the globe such as geology, soil, length of growing season. • The social, economic and political factors affecting food security such as land ownership systems, competition and land grabbing and how these vary from place to place. • Theoretical positions on food security including Malthusian and Boserupian scenarios. • Case study of one place to illustrate how human and physical factors are/have combined to cause issues with food security.

3. What are the threats to global food security?

Key Ideas	Content
3.a. Risks to food security can be identified to highlight the most vulnerable societies.	• Regions, countries and people whose food security is most at risk across the development spectrum. • Why issues related to storage or distribution create geographical pinchpoints where food security is at risk, such as the Suez Canal. • The physical and human causes of desertification and how this changes ecosystems to increase risks to food security. • Case study of one dryland area including: ◦ food security risks and vulnerability are influenced by the specific ecosystem, climate and hydrology ◦ worsening factors such as population change, landgrabbing and climate change.
3.b The food system is vulnerable to shocks that can impact food security.	• How climate change is leading to increasing frequency of extreme weather events such as wild-fire, El-Nino, floods, and drought which can affect food production. • How water scarcity can exacerbate food production issues. • How tectonic hazards can influence food production and distribution. • Case study of one indigenous farming technique in an extreme environment, such as the Arctic, including: ◦ the physical conditions of the environment including ecosystems, terrain and climate. ◦ food production methods used by indigenous people in the environment ◦ threats to the indigenous groups food security.

4. How do food production and security issues impact people and the physical environment?

Key Ideas	Content
4.a. Imbalance in the global food system has physical and human impacts.	• How attempts to increase food production and security can impact the physical environment including: ◦ irrigation and salinisation ◦ deforestation and the impacts on biodiversity ◦ changing landscapes ◦ water quality from agrochemicals. • Case study of how one physical environment is/has been impacted by food production methods including the specific short- and long-term impacts on the environment. • How food security issues impacts people including: ◦ health issues associated with food shortages ◦ health issues associated with food surpluses and poor diet ◦ harmful impacts on human health as a result of the increased use of chemicals and pesticides. • Case studies of two places at contrasting levels of economic development to illustrate the implications of poor food security on the lives of people.

5. Is there hope for the future of food?

Key Ideas	Content
5.a. Food is a geopolitical commodity; a number of key players will continue to influence the global food system.	• The opportunities between countries to ensure food security including: ◦ agricultural trading policies ◦ the role of the World Trade Organization ◦ appropriate aid. • Investigate the role and responsibilities of the following in influencing the global food system: ◦ agribusiness ◦ Trans-National Corporations such as Unilever ◦ food retailers such as Tesco ◦ fair trade organisations.
5.b. There is a spectrum of strategies that exist to ensure and improve food security.	• Approaches to increasing food security can vary from short term relief to capacity building and long term system redesign. • The effectiveness and sustainability of a range of techniques that exist to improve food security from large-scale technological techniques down to small-scale bottom up and appropriate approaches. • Case studies of two contrasting places at different levels of development and the strategies and techniques that have been used to ensure or improve food security.